ETL: 3-5 lesions/leaf early to late tillering or 2-5 neck infected plants/m^2 at panicle initiation to booting.

Spray tebuconazole 25% WG @750 ml/ha or azoxystrobin 18.2% w/w + difenoconazole 11.4% w/w SC @ 0.1% or 1 ml / litre water or tebuconazole 50% + trifloxystrobin 25% WG@200 g/ha: Tricyclazole 75% WP @ 300-400 g/ha may also be applied which is not recommended for export to EU.

Sheath blight

ETL: Lesions of 5-6 mm in length and 2-3 infected plants/ m^2 early to late tillering or 5 infected plants/ m^2 at panicle initiation to booting.

Spray difenoconazole 25% EC @0.05% or 50 ml/100 litre water or difenoconazole 11.4% w/w SC @ 0.1% or 1 ml/litre water or tebuconazole 50% + trifloxystrobin 25%WG@200 g/ha or azoxystrobin 11% + tebuconazole-18.3% SC W/W@ 750 ml/ha.

Bacterial Leaf Blight

ETL: 2-3 infected leaves/ m^2 early to late tillering Spray streptomycin @ 16 g/ha.

VALIDATION OF IPM

Successfully validated IPM in :

i. Pusa Basmati 1 in Haryana and Punjab during *kharif* 2017 and 2018 in 5000 and 10000 ha, respectively.

ii. Taraori Basmati in Panipat (Haryana) during 2002-04 in 160 ha.

iii. Dehraduni Basmati in Dehradun during 2005-07 in 100 ha.

iv. Pusa Basmati 1121 in Sonipat (Haryana) during 2008-10 in 40 ha and Gautam Budg Nagar (Uttar Pradesh) during 2010-18 in 450 ha.

v. Pusa 1509 in Haridwar, (Uttarakhand) during 2017-18 in 60 ha.

IMPACT OF IPM

1. **Impact on cost reduction per ha**: Reduction in cost of inputs along with labour charges by Rs. 4217/- per ha in IPM over non-IPM chemical or Farmers' Practices (FP).
2. **Impact on yield**: Enhancement in paddy yield by 20.28% over FP; Net returns of Rs.18254/- per ha in IPM over FP (Average of 5 years).
3. **Benefit cost ratio (B-C):** Enhancement in B-C ratio in IPM over FP; Average B-C ratio 3.80 in IPM against 2.63 in FP.
4. **Environmental benefits using IPM (specific benefits):**

- Chemical pesticide uses reduced to 0.54 sprays (54.94 g *a.i.*/ha) in IPM against 2.05 sprays (742.84 g *a.i.*/ ha) in FP.
- Conservation of natural enemies: Enhancement in population of spiders, ranged 1.88 to 7.0 adults/hill in IPM against 1.02 to 3.89 adults/ hill in FP.
- Enhancement in population (1.5-2 times) of antagonistic fungi (*Trichoderma* Spp.) and bacteria (*Pseudomonas fluorescens*) in soil.
- Reduction in chemical pesticide residue in grains (below maximum residue level), acceptable in national and international market.

FOR MORE INFORMATION CONTACT
Director, ICAR- National Research Centre for Integrated Pest Management
New Delhi – 110012
Phone: 011-2584 3935, E-mail: director.ncipm@icar.gov.in

NCIPM

INTEGRATED PEST MANAGEMENT IN BASMATI RICE

IPM Strategy with Safe Pesticide Residue Rice Production

भाकृअनुप
ICAR
Developed by
ICAR-National Research Centre for Integrated Pest Management
New Delhi – 110012

INTROUCTION

Basmati rice is cultivated in about 1.6 million ha of north Indian states and fetches high price in domestic and export markets. The notified Basmati varieties mainly comprises of Basmati-370, Basmati-386, Type-3 (Dehraduni), Taraori, Ranbir Basmati, Pusa-1509, Pusa Basmati-1, CSR–30 and Pusa Basmati-1121. The yield potential of these varieties is severely hampered due to substantial losses caused by biotic stresses as there is no inbuilt resistance in them to any of the pests. Extensive surveys of Basmati growing areas by ICAR-NCIPM revealed excessive and injudicious use of pesticides and chemical fertilizers by farmers which aggravated the pest menace, secondary pest outbreaks, residue problems in grains, soil and water, environmental degradation and rejection of many export consignments. The problem to some extent has been successfully managed by integrated pest management (IPM). IPM is a decision support system for integrating eco friendly pest control tactics, aiming to maximize the use of biological control, taking into account the interests of and impacts on the producers, society and the environment and is based on cost/benefit analyses.

Brown Plant Hopper:
Adults (Left) and Hopper Burn (Right)

OBJECTIVES OF IPM

- To create awareness among farmer community about the conservation and augmentation of natural enemies to contain major pests
- To empower the farmers to take decision on need based application of safer pesticides to reduce the injudicious use of hazardous chemical pesticides.
- To help the industry for export of basmati rice by promoting IPM and restricting the farmers for application of recommended pesticides.

IPM MODULE

Enriching soil nutrients

Sowing of green manure, 'Dhaincha'(*Sesbania aculeata*)/Mung.

Burying of 'Dhaincha' after 45-50 DAS/Mung residues after picking of mature pods and puddling of field.

Seed selection, cleaning and treatment

Use of certified seed.

When farmers using own seed it should be cleaned by 2% salt solution for about 15 minutes, discard floating seeds and wash the heavy seeds with fresh water.

Seed treatment with carbendazim 50%WP or thiram 75% WS @ 2g/kg seed.

Sowing time 1-15 June.

Transplanting

Seedling root dipping in *Pseudomonas fluorescens* (liquid formulation 2.0 X 10^8 cfu, 10 ml/litre of water; powder formulation 10 g/litre of water) for 30 minutes Planting 2-3 seedlings/ hill; spacing row to row 20 and hill to hill 15 cm.

Transplanting time 1-15 July.

Fertilizer application

Based on soil testing (reduce fertilizer if green manure has been incorporated).

Nitrogen may be applied in three split doses at 0, 21 and 41 days after transplanting.

Zinc should never be applied with phosphorus fertilizer.

Pest monitoring

Fixing of Yellow Stem Borer (YSB) pheromone traps @ 5 traps/ha for monitoring moth population.

Monitoring of insect pests and diseases by scout at weekly interval.

IPM interventions

Following (Pest wise) interventions are needed when pest reaches at economic threshold level (ETL).

YSB

ETL:2 egg-mass/m^2 or 1 moth/m^2 or 25 moths/trap /week or 10% dead heart.

Release of egg parasitoid, *Trichogramma japonicum* @ 1.5 lakh/ha (affixed as *Tricho* cards) only when egg masses or moths are observed or apply chlorantraniliprole 0.4% GR @10 kg/ha or flubendamide 39.35% EC@50 ml/ha or flubendamide 20% SG@ 125 g/ha; Cartap hydrochloride 4%G @18.7 kg/ha may also be applied (Not recommended for export to EU or US).

Leaf Folder

ETL: Two fully damaged leaves (FDL) with larva/hill or 20% infestation on leaf basis or 1 larva observed per hill.

Release of egg parasitoid, *Trichogramma chilonis* @ 1.5 lakh/ha (affixed as *Tricho* cards) only when egg masses or moths are observed or apply chlorantraniliprole 18.7%SL @150 ml/ha or flubendamide 39.35% EC@ 50 ml/ha or flubendamide 20% SG@ 125 g/ha; Cartap hydrochloride 50% SP @1000 g/ha may also be applied but are not recommended for export to EU or US.

Brown Plant Hopper

ETL: 10-15 hoppers/hill at early to late tillering or 15-20 hoppers/hill at panicle initiation to booting stage.

Drain out water from the field for 2-3 days.

Spray buprofezin 25%SC@ 800 ml/ha or dinotefuran 20% SG@150 g/ha or imidachloprid 70% WG @35mg/ha; Pymetrazine 50%WG @300 g/ha or ethofenprox 10% EC @500 ml/ha may also be applied but are not recommended for export to EU or US.

Blast

ETL: 3-5 lesions/leaf early to late tillering or 2-5 neck infected plants/m^2 at panicle initiation to booting.